Healthcare Financial Software Engineering & Systems Integration

Health Financial Systems | Morgan Briggs

The challenge

Healthcare financial reporting required dependable report access, consistent financial calculations and validated movement of data between web, desktop and Excel workflows.

Role and scope

Software Developer, Health Financial Systems | Approx. 2010/2011 - February 2015. One of two developers on HCRIS, working under the lead programmer. Responsibilities spanned backend reporting, assigned calculation development, application integration, databases and production support.

HCRIS reporting and access

One of two HCRIS developers under the lead programmer; developed C#/ASP.NET backend report generation, historical searches and downloads. Implemented selected Medicare financial calculations from provided specifications; validated outputs against expected results and sample reports. Built database-backed authentication and subscription/facility-based authorization; developed Salesforce customer permissions functionality.

Database engineering and measurable impact

Developed SQL Server/MySQL schemas, queries and stored procedures; optimized databases and supported backups, restores and data validation. Optimized SQL queries and databases, increasing the number of reports available for immediate download by approximately 50%. This measures report availability, rather than a 50% increase in query speed.

Desktop development and Excel integration

Added features and customer enhancements to Delphi Medicare desktop cost-reporting software; debugged and fixed defects. Integrated Excel VBA with Delphi components through a DLL for bidirectional data exchange, calculation execution and validation that blocked invalid imports without opening the desktop UI. For the Excel/API integration, I worked directly with CEO Becky Dolin to jointly define customer needs and requirements. Becky conveyed customer feedback; together we clarified what the solution needed to accomplish. I implemented and iteratively refined the integration with her until it met customer needs.

Imports and operational reliability

Built large-scale scheduled and manual imports using C#, SQL, Delphi and VBA, with validation, logging and failure notifications. Configured Windows Server and IIS, deployed websites and troubleshot production issues; used Visual Studio/TFS, personal testing and lead-programmer review.

Engineering approach

Implemented assigned functionality from specifications, tested changes personally and worked through lead-programmer review. Calculation validation used expected results and sample reports; import validation identified errors and blocked invalid spreadsheet imports.

Technology

C# / ASP.NET / Delphi / VBA / DLL integration / SQL Server / MySQL / Windows Server / IIS / Visual Studio / TFS / Salesforce customer permissions